



Inspiring Excellence

BRAC UNIVERSITY

CSE-350: Digital Electronics and Pulse techniques

Exp-03: Study of a TTL NAND gate with totem pole output

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OBJECTIVES

1. Building standard TTL NAND Gate.
2. Measure the voltages and verify the circuit.

Equipments and component list

Equipments

1. Digital Multimeter
2. DC power supply

Component

- NPN Transistor (C828) - x5 pieces
- Diode 1N4003 - x1 piece
- Capacitor - $4.7 \mu\text{F}$ - x1 piece
- Resistors -
 - ◆ 1K - x1 piece
 - ◆ 1.5K - x1 piece
 - ◆ 1K - x1 piece
 - ◆ 100 - x1 piece

Task-01: TTL NAND gate

THEORY

In this task, we will implement a Transistor-Transistor Logic (TTL) NAND gate with a totem-pole output. Transistor-Transistor Logic, or TTL, refers to the technology for designing and fabricating digital integrated circuits that employ logic gates consisting primarily of bipolar transistors. TTL is the successor of diode-transistor logic (DTL), overcoming the main problem associated with DTL, i.e., lack of speed. TTL provides

Procedure:

1. Connect the circuit as shown in Figure 1.
2. Observe the output for all possible input combinations and fill up table-1.

Data Table

V_A (V)	V_B (V)	V_1 (V)	V_2 (V)	V_3 (V)	V_4 (V)	V_5 (V)	V_6 (V)	V_Y (V)
0	0	0.624	0.624	61.1m	4.90	4.57	4.99	4.23
0	5	0.686	0.802	230.1m	4.60	4.47	4.99	4.20
5	0	0.639	0.638	73.1m	4.88	4.55	4.99	4.21
5	5	1.469	1.473	0.758	0.777	307.3m	4.98	8.0m

Table 1: Table for TTL NAND gate

Dr. Ashar Kotha

Report

Please answer the following questions briefly in the given space.

1. Why is totem-pole output used in place of a passive pull-up resistor?

Ans.

The totem-pole output is used to supply active pull-up and active pull down functionalities. It is used for supplying speed. Unlike passive pull-up resistors, totem pole components are active, allowing faster transitions between high and low states. In contrast to passive pull-up resistors, totem pole has two output stage transistors amplifiers - one CE and the other CE - that can switch more quickly.

2. What is the function of Q3 transistor (phase-splitter)?

Ans.

Using Q3 transistor (phase-splitter) a signal can be divided into several states. It produces a complementary signal which drives the transistors in the totem-pole output. Effective transitioning between high and low output levels is made possible by this.

3. What may happen if diode D_1 is not used in the circuit?
Ans.

D_1 stops Reverse flow of current if in case V becomes $\uparrow$ than $V_6 \rightarrow$ causing circuit to malfunction (something to do with RA Logic)

If the diode D_1 is not used, the transistor Q_5 will be harmed by reverse current flow when output goes from high to low.

4. What is the mode of operation of the Q_5 transistor when output is HIGH?
Ans.

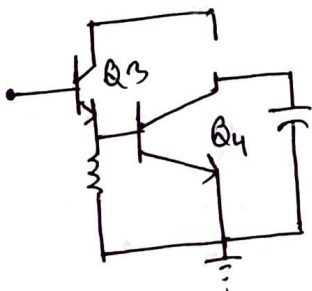
Usually for high output, the Input needs to be low. Therefore, current flows from V_{cc} to V_i which will put Q_3 and Q_4 in cut off mode. Ultimately Q_5 will be in forward active mode.

5. Draw the active portion of the circuit when output is LOW.
Ans.

Output low

$Q_3 \rightarrow ON$

$Q_4 \rightarrow ON$



6. What is the operating mode of Q1 and Q4 transistors when Input A is LOW? Verify using experimental data.
Ans.

Input A $\rightarrow$ Low

Q₁ $\rightarrow$ Saturation mode

Q₄ $\rightarrow$ Cutoff mode

For Q₁

$$V_B = 0.624 = V_1$$

$$V_C = 0.624 = V_2$$

$$V_E = 0$$

$$\therefore V_{BE} = \cancel{0.624} 0.624 - 0 = 0.624$$

$$V_{CE} = 0.624 - 0.624 = 0$$

$$V_{BE}, V_{CE} > 0$$

Therefore, Q₁ is in saturation

For Q₄

$$V_B = 0.0611 = V_3$$

$$V_E = 0$$

$$\therefore V_{BE} = 0.0611 - 0 = 0.06 < 0.5$$

For cutoff mode, we know, $V_{BE} < 0.5$

Therefore, Q₄ is in cut-off mode.

Data Table

V_A (V)	V_B (V)	V_1 (V)	V_2 (V)	V_3 (V)	V_4 (V)	V_5 (V)	V_6 (V)	V_Y (V)
0	0	0.624	0.624	61.1m	4.9	4.57	4.99	4.23
0	5	0.686	0.802	230.1m	4.60	4.42	4.99	4.20
5	0	0.639	0.638	73.1m	4.88	4.55	4.99	4.21
5	5	1.469	1.423	0.258	0.222	307.3m	4.98	8.0m

Table 2: Table for TTL NAND gate


Signature